3.8 Effects on birds A first indication of the effect of cockle fisheries on birds was obtained through comparison of the development of their usage (measured as the number of bird days per year) of areas open and closed for shellfish fishery (Leopold et al., 2003b). These trends must be studied within the trends observed for the Wadden Sea as a whole. For many bird species that feed on the intertidal flats of the Wadden Sea, a change in trend in the annual usage of the flats occurs somewhere at the end of the 1980s, beginning of the 1990s (Figure 47, Figure 48, Figure 49, Figure 50). This corresponds to the disappearance of the intertidal mussel beds. In the graphs we fitted a trend line using splines (Hastie & Tibshirani, 1990) for the entire period, and for the periods before and after the disappearance of the intertidal mussel beds. We subsequently tested whether the change in trend was statistically significant. Changes from a negative trend towards a positive trend outnumber changes from a positive trend towards a negative trend. For species feeding either primarily on worms or shellfish, the greater the percentage shellfish in the diet, the larger the decline during the last ten years (van Roomen et al., 2004). The only species not fitting the trend is the avocet, which feeds on worms, yet showed a decline in recent years. For species that feed primarily on prey other than bivalves or worms (Figure 50), and for species whose diet is a mixture of shellfish, worms and other prey (Figure 49), no simple patterns emerge. Four species of birds were identified as shellfish eaters: common eider, oystercatcher, herring gull and knot. All these species have declined in recent years, although the onset of the decline varies between the species (Figure 47). Of these, the common eider and the oystercatcher feed on large shellfish, whereas knot and herring gull feed on small shellfish. Since common eiders depend primarily on the subtidal areas for their food supply, we did not investigate the effect of closing intertidal areas. In contrast to waders, several counts of eiders covering the entire Wadden Sea are available for the period before 1975. We did not include these counts in (Figure 47) to facilitate comparison with the other bird species. However, including these counts makes clear that the suggestion that eiders have continually decreased from 1975 onwards is an artefact of this selection of the data. Eiders have increased during the 1960s, fluctuated in numbers during the 1970s and 1980s, and declined during the 1990s. The recent decline in the Wadden Sea was accompanied by an increase in the number of eiders using the North Sea coastal zone (see Figure 72 in chapter 5), but this increase did not fully compensate for the decline in the Wadden Sea (Ens & Kats, 2004). The interpretation of these changes is discussed in chapter 5, where the policy of food reservation for common eiders is evaluated. The numbers of oystercatchers declined after about 1990 (Figure 47). The decline in the number of oystercatchers started in the first half of the 1990s. Since the strong and sudden decline due to the severe winter of 1996/1997, no further decline has occurred, nor have numbers recovered. The lowered numbers are largely due to the disappearance of the intertidal mussel beds in 1990 (Smit et al., 1998; Rappoldt et al., 2003a), see also chapter 5. The fact that we did not see a sudden decline in the number of oystercatchers following the rather sudden disappearance of the intertidal 